

LECTURE 6

GROUP AND RING THEORY

5. SIMPLE MODULES, COMPLETELY REDUCIBLE MODULES AND FREE MODULES

5.1. Simple modules. Analogous to simple groups, we have the following important concept. Note that here whenever we refer to the unity, we mean non-zero unity.

Definition 5.1. Let R be a ring with (non-zero) unity. An R -module M is called *simple* (or *irreducible*) if $M \neq 0$ and the only R -submodules of M are 0 and M .

Example 5.2. A trivial example of a simple module is a field or a division ring R regarded as a module over itself. In fact, a ring R (with unity) that is not a division ring gives rise to an R -module that is not simple. Indeed, this follows from that fact that a ring R has only (0) and R as its left ideals if and only if R is a division ring.

Example 5.3. Let R be a ring with unity. A minimal left ideal in a ring R is also clearly a simple R -module.

Example 5.4. Let $R = M_n(F)$ be the matrix ring over a field F , and for $1 \leq i, k \leq n$, denote by E_{ik} the matrix with 1 at the (i, k) entry and 0 elsewhere. Then $M = RE_{kk}$ is a simple R -module.

Solution: Let $N \neq 0$ be an R -submodule of RE_{kk} . Note that M is the set of matrices all of whose entries are zero, except possibly in the k th column. Hence a non-zero element of N can be expressed as $\sum_{i=1}^n a_{ik} E_{ik}$ for $a_{ik} \in F$.

Consider $0 \neq \sum_{i=1}^n a_{ik} E_{ik} \in N$. Then for some ℓ , with $1 \leq \ell \leq n$, we have $a_{\ell k} \neq 0$. So $a_{\ell k}^{-1} \in F$ exists. This gives

$$E_{kk} = (a_{\ell k}^{-1} E_{k\ell}) \left(\sum_{i=1}^n a_{ik} E_{ik} \right) \in N,$$

as

$$E_{k\ell} E_{ik} = \begin{cases} 0 & \text{if } i \neq \ell, \\ E_{kk} & \text{if } i = \ell. \end{cases}$$

Hence $RE_{kk} \subseteq N \subseteq RE_{kk}$ giving $N = RE_{kk}$. This proves that M is simple.

We now prove the following characterisation of simple modules.

Theorem 5.5. Let R be a ring with unity, and let M be an R -module. Then the following statements are equivalent:

- (i) M is simple;
- (ii) $M \neq 0$, and M is generated by any $0 \neq x \in M$;
- (iii) $M \cong R/I$, where I is a maximal left ideal of R .

Proof. (i) $\implies$ (ii): Let $0 \neq x \in M$. Then $(x) = Rx$ is a non-zero R -submodule generated by x . So $M = (x)$.

(ii) $\implies$ (iii): As $M \neq 0$, we can take $x \in M \setminus \{0\}$. Then $Rx = M$, and the mapping

$$\begin{aligned} f : R &\rightarrow Rx \\ r &\mapsto rx \end{aligned}$$

is an R -homomorphism of R (regarded as an R -module) to M . The mapping f is clearly surjective. Next let $I = \ker f$. Then by the First Isomorphism Theorem, we have $R/I \cong Rx = M$.

Now let J be a left ideal of R with $I \subsetneq J \subseteq R$. Let $y \in J \setminus I$. Then $f(y) \neq 0$, which gives $Rf(y) = M$. We claim that $R = Ry + \ker f$. Indeed, let $r \in R$. Then $f(r) \in M = Rf(y)$, so we have $f(r) = kf(y)$ for some $k \in R$. Then $f(r) = f(ky)$. Note that $f(r - ky) = f(r) - f(ky) = 0$, and so $r - ky \in \ker f = I$. Hence $r \in ky + I \subseteq Ry + I$. As r was arbitrary, we obtain $R = Ry + I$. Now we have

$$R = Ry + I \subseteq J,$$

hence $J = R$, so I is a maximal left ideal. [Alternatively, since we are assuming (ii), from $M \cong R/I$, as $y + I \neq 0 + I$, the zero element in R/I , and as M is generated by any non-trivial element, this property translates over to R/I . So R/I is generated by any non-trivial element. As $y + I$ is non-trivial, we get immediately that $R(y + I) = R/I$. Hence $J = R$ by the Correspondence Theorem.]

(iii) $\implies$ (i): Let $0 \neq N \subseteq M$ be a submodule. Under the isomorphism $R/I \cong M$, we have by Theorem 4.31 that there exists $I \subsetneq J \subseteq R$ such that $J/I \cong N$. Since I is maximal, we have that $J = R$ which implies that $N = M$, as required. $\square$

One important fact about simple modules is the following result.

Theorem 5.6 (Schur's Lemma). *Let M be a simple R -module. Then $\text{Hom}_R(M, M)$ is a division ring.*

Proof. Let $0 \neq f \in \text{Hom}_R(M, M)$. Consider the R -submodules $\ker f$ and $\text{im } f$ of M . If $\ker f = M$, then $f = 0$, a contradiction. Thus $\ker f = (0)$, so f is injective. Further, if $\text{im } f = (0)$, then $f = 0$, again a contradiction. Therefore $\text{im } f = M$, which implies f is surjective. Hence f is bijective, which proves that f is invertible.

Lastly, we claim that $f^{-1} \in \text{Hom}_R(M, M)$. Indeed, for $a, b \in R$ and $x, y \in M$, we have

$$f(f^{-1}(ax + by)) = ax + by,$$

and

$$f(af^{-1}(x) + bf^{-1}(y)) = af(f^{-1}(x)) + bf(f^{-1}(y)) = ax + by.$$

As f is injective, it follows that $f^{-1}(ax + by) = af^{-1}(x) + bf^{-1}(y)$. Hence the claim. $\square$

5.2. Completely reducible modules. Here we define a more general family of modules that contain simple modules.

Definition 5.7. Let R be a ring with unity. An R -module M is called *completely reducible* (or *semisimple*) if $M = \sum_{\alpha \in \Lambda} M_\alpha$, where M_α are simple R -submodules.

Example 5.8. The vector space $\mathbb{R} \oplus \mathbb{R}$ is completely reducible, since $\mathbb{R}$, being a field, is a simple $\mathbb{R}$ -submodule. The direct sum $C_p \oplus C_q$, of cyclic groups of distinct prime order, is completely reducible.

We now show that in a completely reducible module M , every submodule has a complement.

Theorem 5.9. Let $M = \sum_{\alpha \in \Lambda} M_\alpha$ be a sum of simple R -submodules M_α , and let K be a submodule of M . Then there exists a subset Λ' of Λ such that $\sum_{\alpha \in \Lambda'} M_\alpha$ is a direct sum, and $M = K \oplus (\oplus_{\alpha \in \Lambda'} M_\alpha)$.

Proof. Let $S = \{A \subseteq \Lambda \mid \sum_{\alpha \in A} M_\alpha \text{ is a direct sum, and } K \cap \sum_{\alpha \in A} M_\alpha = (0)\}$. Then $S \neq \emptyset$ since $\emptyset \in S$. (Note, by $\sum_{\alpha \in A} M_\alpha$ we mean (0) when A is an empty set.) The set S is partially ordered by inclusion, and every chain (A_i) in S has an upper bound $\cup A_i$ in S . Thus, by Zorn's Lemma, the set S has a maximal member, say A .

Let $N = K \oplus (\oplus_{\alpha \in A} M_\alpha)$. We claim $N = M$. Let $\beta \in \Lambda$. Since M_β is simple, either $M_\beta \cap N = (0)$ or $M_\beta \cap N = M_\beta$. Now $M_\beta \cap N = (0)$ implies $M_\beta \cap \oplus_{\alpha \in A} M_\alpha = (0)$, so $\sum_{\alpha \in A \cup \{\beta\}} M_\alpha$ is a direct sum (cf. Theorem 4.22), and it has zero intersection with K . This contradicts the maximality of A , therefore, for all $\beta \in \Lambda$ we have $M_\beta \subseteq N$, proving that $N = M$.

[Alternatively: Let $S = \{A \subseteq \Lambda \mid K + \sum_{\alpha \in A} M_\alpha \text{ is a direct sum}\}$, and then the end of the proof simplifies slightly.] $\square$

By choosing $K = (0)$ in Theorem 5.9, we obtain:

Corollary 5.10. If $M = \sum_{\alpha \in \Lambda} M_\alpha$ is the sum of a family of simple R -modules $(M_\alpha)_{\alpha \in \Lambda}$, then there exists a subfamily $(M_\alpha)_{\alpha \in \Lambda' \subseteq \Lambda}$ such that $M = \oplus_{\alpha \in \Lambda'} M_\alpha$.

As a consequence of the above results, we can show that every non-zero submodule and every non-zero quotient module of a completely reducible module is completely reducible (see Exercise Sheet 6).

5.3. Free modules. Here we consider modules that are more similar to vector spaces. Throughout this section, unless otherwise stated, the ring R is a non-zero ring with unity.

Definition 5.11. A finite sequence $x_1, \dots, x_n$ of elements of an R -module M is called *linearly independent* if, for any $a_1, \dots, a_n \in R$, the equation $\sum_{i=1}^n a_i x_i = 0$ implies $a_1 = a_2 = \dots = a_n = 0$. A finite sequence is called *linearly dependent* if it is not linearly independent.

A subset S of an R -module M is called *linearly independent* if every finite sequence of distinct elements of S is linearly independent. Otherwise S is called *linearly dependent*.

Example 5.12. A linearly independent set in the F -module $F[x]$ is $\{1, x, x^2, x^3, \dots\}$. On the other hand, the set $\{1, x, 1+x, x^2\}$ is a linearly dependent set.

Example 5.13. In the R -module $M = R^n$, consider $\{e_1, e_2, \dots, e_n\}$, where e_i is the n -tuple in which all components except the i th are zero and the i th component is 1. This set $\{e_1, e_2, \dots, e_n\}$ is a linearly independent set in R^n .

Definition 5.14. A subset B of an R -module M is called a *basis* if

- (i) M is generated by B ;
- (ii) B is a linearly independent set.

Example 5.15. The basis $\{e_1, e_2, \dots, e_n\}$ of R^n that we have described is called the *standard basis* of R^n .

Unlike vector spaces, not every module has a basis:

Example 5.16. Let G be a cyclic group and regard it as a $\mathbb{Z}$ -module. Then G has a basis if and only if G is infinite. For, let $G = \langle a \rangle$ and let ma , for $m \in \mathbb{Z}$, be some basis element. Then $\lambda(ma) = 0$, for $\lambda \in \mathbb{Z}$, must imply $\lambda = 0$. However, if G is a finite group of order n ,

then $n(ma) = 0$, a contradiction. Hence G must be infinite if it has a basis. Now if $G = \langle a \rangle$ is an infinite cyclic group, then $\{a\}$ is clearly a basis of G as a $\mathbb{Z}$ -module. This proves our assertion.